Empirical notes on EchoKey operator assignments to the Collatz transfer-operator setting

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https://github.com/JGPTech/Fun

Abstract. These notes record a structural-consistency check between two separate frameworks on a single test bed. The frameworks are the EchoKey v2 specification [7], which fixes seven operators on a separable Hilbert space (Cyclicity, Recursion, Fractality, Regression, Synergy, Refraction, Outliers), and Kim's arithmetic Chern–Simons theory [5], in which the open question Q2 of the companion notes [9] asks for a $\mathbb{Z}_p$-valued lift of the scalar observable $H(\gamma_n)$ on Collatz trajectories. For each EchoKey operator we (i) assign it a specific object in the Collatz transfer-operator setting, (ii) compute a numerical prediction from the operator's formal definition, and (iii) compare to a measured anchor from [9]. All seven assignments are consistent with their corresponding anchors within pre-declared tolerances, with five involving independent computations and two serving as internal consistency checks on the recipe. The Outliers test is the only one performing cross-observable recovery: it independently locates the four-piece crossing at step $k^* = 76$ from the v_2-mean trajectory alone, against the measured value of step 75. Nothing here is a theorem in Kim's framework. The seven assignments are offered as a hypothesis-generating decomposition; whether any of them survives a precise lift is open.

These notes assume familiarity with the companion notes [9] (the observable H, the supermartingale conjecture, the four-piece crossing) and the EchoKey v2 specification [7] (the seven operators and their formal definitions). No background in arithmetic topology is required.

1 Setup and framing

1.1 The open question being probed

In [9] the observable

$$H(\gamma_n) = \sum_{j=0}^{|\gamma_n|-1} v_2\big(\gamma_n^{(j)}\big)$$

was introduced on finite Collatz trajectories. Question Q2 of those notes asks for a structural lift of H to a $\mathbb{Z}_p$-valued invariant in Kim's arithmetic Chern–Simons framework [5], specifying (i) a basepoint in $\pi_1^{\mathrm{un}}(\mathrm{Spec}(\mathbb{Z}_2))$, (ii) a target p-adic Lie group, and (iii) the sense in which discrete Collatz steps define morphisms in the étale fundamental groupoid, in such a way that the lift reduces to H in the abelian quotient and is consistent with the empirical anchors of [9] (attractor value $H_{\mathrm{norm}}(\{1,2\}) = 1/3$, tail value $\approx 1.18 \pm 0.18$, supermartingale signature for $\mathbb{E}_{\mu_k}[f]$, four-piece crossing at 96.2% of collapse time with $\Delta\bar{v}_2 = -0.48$).

These notes do not solve Q2 in Kim's framework. They instead take a separate operator framework — the EchoKey v2 specification [7] — and decompose the structure that Q2 demands into seven operator slots, then test whether the decomposition is consistent with every anchor reported in [9].

1.2 What the test does and does not show

A successful seven-out-of-seven match shows that EchoKey's operators, applied via the assignments of Section 3, do not contradict any anchor of [9] and reproduce the empirical anchors of Q2 within tolerance. It does not show that EchoKey's operators *are* the correct decomposition of Kim's framework, and it is not a proof inside Kim's framework. The whole exercise should be read as hypothesis-generating: the assignments below are the natural targets of each operator's formal definition, not forced ones; other assignments are possible, and several of the present ones may be wrong or incomplete. The upgrade from this consistency check to a Kim-internal proof is recorded as OP-EK1 in Section 6.

1.3 Frameworks in use

The arithmetic-topology background is the M^2KR dictionary [1, 4, 2, 3], with Kim's arithmetic Chern–Simons program [5] as the active formalization layer. The Collatz-side empirical setup is the transfer-operator framework of [9]. The EchoKey v2 specification is [7]; the applied EchoKey-EFECGSC work [8] provides the simulation context in which six of the seven operators were first deployed, with Refraction taken from the v2 specification proper.

2 The EchoKey v2 operators

We collect the seven operator definitions used here, in the notation of [7]. Throughout, $\mathcal{H}$ is a separable Hilbert space and $\Psi \in \mathcal{H}$ a state. (Note that the symbol $\mathcal{F}$ appears in two roles in the spec: as the generic map being iterated in the Recursion definition, and as the Fractality operator itself; both usages are retained here for fidelity with [7].)

Definition 1 (Cyclicity [7], Def. 3.1)**.** *For period $T > 0$ and angular frequencies $\omega_n = 2\pi n/T$,*

$$\mathcal{C}_T[f](t) = \sum_{n\in\mathbb{Z}} \left(\frac{1}{T} \int_0^T f(\tau) e^{-i\omega_n \tau}\, d\tau \right) e^{i\omega_n t}.$$

$\mathcal{C}_T$ is the Fourier-like projector onto the period-T subspace of $L^2([0,T])$.

Definition 2 (Recursion [7], Def. 4.1)**.** *For a map $\mathcal{F} : \mathcal{H} \to \mathcal{H}$, set $\mathcal{R}^{(n)}[f] = \mathcal{F}(\mathcal{R}^{(n-1)}[f])$ with $\mathcal{R}^{(0)}[f] = f$, and $\mathcal{R}[f] = \lim_{n\to\infty} \mathcal{R}^{(n)}[f]$ when the limit exists. Existence follows from Banach contraction whenever $\mathcal{F}$ is a contraction on a metric subset of $\mathcal{H}$.*

Definition 3 (Fractality [7], Def. 5.1)**.** *For a dilation $D_\lambda \phi(x) = \phi(\lambda x)$ with $\lambda > 1$, weight sequence $\{\alpha_k\}_{k\geq 0}$, and a map $\mathcal{E}$ on $\mathcal{H}$,*

$$\mathcal{F}[\mathcal{E}](\Psi) = \sum_{k=0}^{\infty} \alpha_k\, \mathcal{E}(D_{\lambda^k}\Psi).$$

Definition 4 (Regression [7], Def. 6.1)**.** *For a reference state $\Psi_\star \in \mathcal{H}$ and rate $\lambda > 0$,*

$$\mathcal{G}_\lambda[\Psi](t) = \Psi_\star + e^{-\lambda t}\big(\Psi(0) - \Psi_\star\big).$$

Definition 5 (Synergy [7], Def. 7.1)**.** *For components $\Psi = (\Psi_1, \ldots, \Psi_n) \in \mathcal{H}^n$, coefficients κ_{ij}, and a bilinear form B,*

$$\mathcal{S}[\Psi] = \sum_{1\leq i<j\leq n} \kappa_{ij}\, B(\Psi_i, \Psi_j).$$

Definition 6 (Refraction [7], Def. 8.1)**.** *For a layer index $L \in \mathbb{N}$ and a transform density $\eta : \mathcal{H} \to \mathcal{H}$,*

$$\Pi[\Psi; L] = \Psi + \mu L\, \eta(\Psi),$$

interpreted as a small deformation across an interface or domain change.

Definition 7 (Outliers [7], Def. 9.1)**.** *For times and weights* $\{(t_k, w_k)\}_{k\in\mathbb{N}}$ *and jump directions* $\Xi_k \in \mathcal{H}$*,*

$$\mathcal{O}[\Psi](t) = \Psi(t) + \sum_k w_k\, \delta(t - t_k)\, \Xi_k.$$

3 Operator assignments

For each operator we record below the Collatz/transfer-operator object to which we assign it, together with the empirical anchor from [9] against which the assignment will be tested. The table is offered as a hypothesis-generating decomposition; several rows are likely incomplete or wrong, and the assignments are recorded in advance of the test rather than chosen to make it pass.

Op	**Name**	**Assigned object**	**Anchor from [9]**
$\mathcal{C}$	Cyclicity	Attractor cycle $\{1,2\}$ realised as $[1,2,1]$	$H_{\text{norm}}(\{1,2\}) = 1/3$
$\mathcal{R}$	Recursion	Iterated transfer operator $(T_3)_*$ on μ_k	Tail-H std contraction rate $\alpha < 1$ (Banach condition)
$\mathcal{F}$	Fractality	Multiscale $\mathbb{Z}_2 = \varprojlim_n \mathbb{Z}/2^n$, $\lambda = 2$, $\alpha_k = 2^{-k}$	Pearson $r(H, \lvert\gamma\rvert) = 0.96$
$\mathcal{G}$	Regression	Cesàro mean $C_k = \frac{1}{k+1}\sum_{j\le k} \mathbb{E}_{\mu_j}[f]$	Steady state $f_\star \in [0.4, 0.8]$ and decay rate $\lambda > 0$
$\mathcal{S}$	Synergy	Parity-branch bilinear coupling $B(\Psi_{\text{odd}}, \Psi_{\text{even}})$ at the four-piece crossing	$M_1 = \lvert\mathbb{E}[f\lvert B{=}1] - \mathbb{E}[f\lvert B{=}0]\rvert = 0.625$
Π	Refraction	Siegel (p,q)-adic conjugacy *recast through the EchoKey Refraction lens* as $L = 1$ layer transform	v_2-mean invariance: max layerwise difference $= 0$
$\mathcal{O}$	Outliers	Four-piece crossing as singular impulse in v_2-mean	Impulse location k^* and amplitude $\Delta\bar{v}_2 = -0.48$

4 Verification results

All numbers in this section are produced by `collatz_echokey_probe.py` (CC0, in the same repository), which imports `collatz_arithmetic_probe.py` as the source of anchor data.

4.1 Cyclicity: attractor signature

The attractor cycle $1 \to 2 \to 1$ realised as $[1,2,1]$ has v_2-sequence $[0,1,0]$. The discrete Fourier projector onto period-3 sequences is the identity on length-3 vectors; idempotence is satisfied to floating-point precision ($\|\mathcal{C}_T^2 x - \mathcal{C}_T x\| = 7.4\times 10^{-17}$). The substantive prediction is the time average of v_2 over one period:

$$\frac{1}{3}\big(v_2(1) + v_2(2) + v_2(1)\big) = \frac{1}{3} = H_{\text{norm}}(\{1,2\}).$$

Predicted 0.333, measured 0.333; relative error 0. **Match.**

4.2 Recursion: long-run contraction rate

The iterated transfer operator $(T_3)_*$ acting on the support measure produces a sequence of supports $S_0, S_1, \ldots$ with corresponding tail-H distributions over the last K steps. The companion notes report standard deviations $\sigma(K = 5) = 0.395$, $\sigma(K = 10) = 0.593$, $\sigma(K = 20) = 0.180$. The non-monotone behaviour from $K = 5$ to $K = 10$ reflects a transient regime in which short trajectories dominate small K; the genuine contraction emerges once K is comparable to typical trajectory length. Using the values at $K = 10$ and $K = 20$ to estimate a local exponential rate gives a per-step contraction

$$\alpha = 0.887 < 1,$$

which satisfies the Banach contraction condition under which $\mathcal{R}[f] = \lim_n \mathcal{R}^{(n)}[f]$ exists. Predicted $\alpha < 1$, measured $\alpha = 0.887$. **Match** (threshold test). The estimate uses only two data points, so this is the weakest of the matches; a denser sweep is recorded as OP-EK2.

4.3 Fractality: extensivity of H

The multiscale structure $\mathbb{Z}_2 = \varprojlim_n \mathbb{Z}/2^n$ with weights $\alpha_k = 2^{-k}$ suggests that a well-converged $\mathcal{F}$-sum is dominated by its extensive component, giving Pearson correlation $r \gtrsim 0.9$ between H and trajectory length. The companion notes report $r = 0.96$ over $n \leq 100$. Predicted $r \gtrsim 0.9$, measured $r = 0.96$. **Match.**

The fitted slope $H \approx 0.589\,|\gamma_n| + 7.523$ is below what a naive uniform-bit heuristic might suggest: the 2-adic digit distribution on $\mathbb{Z}_2^\times$ (geometric with parameter 1/2) has mean 1. The gap reflects that Collatz trajectories concentrate disproportionately on odd nodes (where $v_2 = 0$), suppressing the empirical per-step mean of v_2 below the uniform-digit value. The match is on the qualitative correlation, not on the slope; resolving the slope quantitatively is recorded as OP-EK3.

4.4 Regression: Cesàro relaxation

The supermartingale conjecture of [9] concerns the Cesàro mean $C_k = \frac{1}{k+1}\sum_{j\leq k} \mathbb{E}_{\mu_j}[f]$. Fitting $C_k = f_\star + e^{-\lambda k}(C_0 - f_\star)$ for $N = 200$ over the full collapse trajectory:

$$C_0 = 0.667, \quad C_{\text{final}} = 0.661, \quad f_\star = 0.610, \quad \lambda = 0.0011 \text{ per step.}$$

Both prediction conditions hold: $f_\star \in [0.4, 0.8]$ (the vacuum $\mathbb{E}[f]$ range as set up in [9], bounded by attractor values $|1|_2 = 1$ and $|2|_2 = 0.5$) and $\lambda > 0$ (genuine relaxation). **Match.**

4.5 Synergy: parity-branch coupling at the crossing

At the four-piece crossing for $N = 200$, the support is $\{1, 2, 4, 5\}$ with conditional expectations $\mathbb{E}[f \mid B{=}1] = 1.000$ (odd nodes: $\{1, 5\}$) and $\mathbb{E}[f \mid B{=}0] = 0.375$ (even nodes: $\{2, 4\}$). The Synergy bilinear pairing $B(\Psi_{\text{odd}}, \Psi_{\text{even}})$ taken as the absolute difference of conditional expectations evaluates to $|1.000 - 0.375| = 0.625$, equal to the M_1 vacuum measure at the crossing. Predicted 0.625, measured 0.625. **Match.**

This test is near-tautological: the Synergy recipe and the M_1 definition produce the same arithmetic. What it verifies is that the prescribed coupling form lands on the right anchor, not that it generates new information. A non-tautological version (a higher-order pairing producing a quantity not directly defined by the recipe) is recorded as OP-EK4.

4.6 Refraction: Siegel conjugacy as layer transform

Siegel's (p, q)-adic conjugacy [6] identifies T_3 on $\mathbb{Z}_2$ with the 2-adic shift σ. We recast this conjugacy through the EchoKey Refraction lens as the $L = 1$ layer transform $\Pi[\Psi; 1] = \Psi + \mu\eta(\Psi)$; this is a structural reinterpretation for the purposes of the present analysis, not a theorem of the (p, q)-adic spectral framework itself. On that reading, the conjugacy preserves the 2-adic valuation: for any integer n, $v_2(n)$ computed by direct integer arithmetic equals $v_2(n)$ computed as the count of trailing zeros in the base-2 expansion. The per-trajectory v_2-mean computed in the integer layer and the bit-string layer was compared over $n = 1, \ldots, 100$; the maximum difference across all trajectories is 0 (exactly, not just within tolerance). **Match.**

This is a structural identity: the integer and bit-string definitions of v_2 agree by construction. The test verifies that the Refraction operator, applied to the Siegel conjugacy on this reading, preserves the invariant the conjugacy preserves. Like Synergy, this is a consistency check rather than a non-trivial computation; a non-abelian extension (recovering H itself rather than only its mean) is recorded as OP-EK5.

4.7 Outliers: independent recovery of the crossing

The Outliers operator models the four-piece crossing as a singular impulse in the v_2-mean trajectory. The pipeline:

1. Compute v_2-mean(k) for each step k in the $N = 200$ collapse trajectory.
2. Compute the windowed mean $\overline{v_2}_W(k)$ with $W = 5$.
3. Locate $k^* = \arg\min_k \big(\overline{v_2}_W(k+1) - \overline{v_2}_W(k)\big)$, the step with the largest negative jump.
4. At the located k^*, compute the symmetric pre/post-window means and report their difference as $\Delta\bar{v}_2$.

The pipeline is given no information about support size or crossing criteria, only the v_2-mean trajectory. The result:

$$k^* = 76, \qquad \Delta\bar{v}_2 = -0.557.$$

The companion notes report the four-piece crossing at $k = 75$ (defined by support shrinkage to size ≤ 4) with $\Delta\bar{v}_2 = -0.480$. The location is recovered to within one step; the amplitudes differ by 0.077, or 16% relative to the measured value. Predicted -0.557, measured -0.480 (tolerance 20%). **Match.**

We note that the pipeline locates a *larger* drop than the support-based measurement, so the agreement is qualitative rather than tight: whether the pipeline is detecting exactly the same event or a slightly displaced feature (e.g., a multi-step drop that straddles the support-based definition) is not settled here. What the test does establish is that a jump-detection pipeline running only on $\bar{v}_2$ finds a singular event at essentially the same location as the support-based criterion identifies. This is the only test performing cross-observable recovery, and therefore the most structurally informative of the seven; the kind of cross-method agreement that distinguishes a structural assignment from a relabeling.

4.8 Summary table

Op	Predicted	Measured	Tol.	Verdict
$\mathcal{C}$	0.333	0.333	10^{-6}	MATCH
$\mathcal{R}$	$\alpha < 1$	$\alpha = 0.887$	threshold	MATCH
$\mathcal{F}$	$r \gtrsim 0.9$	$r = 0.96$	threshold	MATCH
$\mathcal{G}$	$f_\star \in [0.4, 0.8]$, $\lambda > 0$	$f_\star = 0.610$, $\lambda = 0.0011$	threshold	MATCH
$\mathcal{S}$	0.625	0.625	10^{-6}	MATCH
Π	0	0	10^{-12}	MATCH
$\mathcal{O}$	-0.557	-0.480	20%	MATCH

Seven of seven assignments produce predictions matching their anchor within pre-declared tolerances. Two of the seven $(\mathcal{S}, \Pi)$ are consistency checks on the recipe rather than independent computations; the remaining five involve independent computations that reconstruct or match an anchor of [9]. The Outliers test is the one performing genuine cross-method recovery.

5 How the seven slots map to Q2's three requirements

Q2 of [9] asks for three structural objects. The seven-operator decomposition lines up with these as follows. We emphasise that the alignment is interpretive — the operators *can be* read this way, not that they *must be*.

(i) **The basepoint in $\pi_1^{\mathrm{un}}(\mathrm{Spec}(\mathbb{Z}_2))$.** The Cyclicity assignment selects the attractor cycle $\{1, 2\}$ as the period-3 realisation $[1, 2, 1]$. The basepoint is the fixed state of $\mathcal{C}_T$ acting on the iterate sequence: in operator terms, the unique state on which the period projector acts as the identity. Empirically this state has $H_{\mathrm{norm}} = 1/3$ and the canonical signature $M_4 = 2$ from [9].

(ii) **The target p-adic Lie group.** The Recursion and Fractality assignments together generate the multiscale structure: $\mathcal{R}$ provides iterative self-similarity (with measured contraction rate $\alpha = 0.887$), and $\mathcal{F}$ provides the layered scaling $\alpha_k = 2^{-k}$ realising the $\mathbb{Z}_2 = \varprojlim_n \mathbb{Z}/2^n$ tower. A natural candidate target group consistent with these assignments is the abelian quotient of the unipotent radical of $\mathrm{GL}_n(\mathbb{Z}_2)$ generated by the layered shifts; specifying the precise non-abelian extension is part of Q6 of [9].

(iii) **The groupoid morphisms.** The Synergy and Refraction assignments together cover the morphism content: $\mathcal{S}$ couples adjacent parity branches (verified by the M_1 recovery), and Π provides the layer transform between the integer and shift representations (verified by exact v_2-mean invariance). A discrete Collatz step $T_3 : n \mapsto T_3(n)$ would, on this reading, factor through both: $\mathcal{S}$ governs the within-layer parity transition, Π governs the between-layer identification.

The Regression and Outliers operators do not address Q2's three requirements directly. They certify dynamical properties any honest holonomy lift would need to respect: $\mathcal{G}$ ensures supermartingale convergence (a prerequisite for any optional-stopping argument on the lifted invariant), and $\mathcal{O}$ identifies the four-piece crossing as a singular impulse rather than a smooth transition, constraining the regularity class of the lift.

What this section does and does not do: it lines up the seven operator slots with Q2's three requirements in a way consistent with the empirical anchors. It does not show that the seven-operator decomposition coincides with the étale-fundamental-groupoid structure on $\mathrm{Spec}(\mathbb{Z}_2) \setminus S$

for any particular ramification set S. That coincidence, if it holds, would be a theorem in Kim's framework; it is not one that these notes contain.

6 Open questions

Tagged *[analytic]* for pen-and-paper proof, *[structural]* for identification of an object in an existing framework, *[computational]* for an extension of the present probe.

OP-EK1. *[structural]* **Internal proof in Kim's framework.** The seven-operator decomposition is consistent with every anchor of [9], but it is not currently a theorem in Kim's arithmetic Chern–Simons framework. Show that the EchoKey assignments correspond to specific structures in the étale fundamental groupoid $\pi_1^{\mathrm{un}}(\mathrm{Spec}(\mathbb{Z}_2) \setminus S, b)$ for some basepoint b and ramification set S, upgrading structural consistency to a Kim-internal statement.

OP-EK2. *[computational]* **Strengthen the Recursion test.** The contraction rate $\alpha = 0.887$ is a two-point fit ($K = 10$ and $K = 20$). Compute tail-H std at intermediate values $K \in \{12, 14, 16, 18\}$ and confirm geometric decay across the full range, or document where it fails.

OP-EK3. *[structural]* **Resolve the Fractality slope discrepancy.** The fitted slope of H versus length is 0.589, below the uniform-digit heuristic value of 1. Derive the corrected expected slope from the empirical odd/even visit ratio along Collatz trajectories and compare.

OP-EK4. *[computational]* **Non-tautological Synergy test.** The current Synergy test reproduces M_1 by construction. Replace the bilinear pairing with a higher-order pairing (an m-ary term in the sense of Question 7.2 of [7]) that predicts an anchor not directly defined by the recipe.

OP-EK5. *[structural]* **Non-abelian Refraction.** The current Refraction test verifies an abelian invariant (v_2-mean). Determine the correct non-abelian extension required to recover H itself (rather than only its mean) under the Siegel layer transform.

OP-EK6. *[structural]* **Operator-ordering sensitivity.** Question 10.2 of [7] asks how sensitive the v2 composite $\mathcal{H}_{v2} = \mathcal{C} \circ \mathcal{R} \circ \mathcal{F} \circ \mathcal{G} \circ \mathcal{S} \circ \Pi \circ \mathcal{O}$ is to operator ordering. Determine which pairs commute on the Collatz test bed and which produce ordering-dependent invariants.

OP-EK7. *[structural]* **Lift to general $\mathbb{Z}_p$.** Extend the analysis from $\mathbb{Z}_2$ to general $\mathbb{Z}_p$ and from the shortened Collatz map to general dynamical systems on local fields. The seven assignments here are specific to $p = 2$; identifying which are universal and which are 2-adic accidents is a prerequisite for any further extension of the framework.

OP-EK8. *[computational]* **Tighten the Outliers recovery.** The jump-detection pipeline locates a larger drop (-0.557) than the support-based measurement (-0.480) at a slightly displaced step ($k^* = 76$ vs. $k = 75$). Determine whether the discrepancy reflects a

multi-step drop straddling the support-based definition, a window-size artefact, or a genuine feature of the $\bar{v}_2$ trajectory distinct from the support-size criterion.

7 Summary

What these notes contain: seven operator assignments (Section 3) and a verification record (Section 4) showing all seven match anchors from [9] within pre-declared tolerances. Five of the seven involve non-trivial computations; two are consistency checks. The strongest test is Outliers, which independently recovers the four-piece crossing location at $k^* = 76$ from the v_2-mean alone, against the measured value of step 75.

What these notes do not contain: any internal proof in Kim's framework. The seven-operator decomposition lines up with Q2's three requirements (basepoint, target group, groupoid morphisms) in a way consistent with every anchor, but the alignment is interpretive, not theorematic. Whether the alignment survives a precise lift is OP-EK1; the other open questions in Section 6 are the obvious places where the present analysis is thinnest.

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